

Fluid Highlights: Stylized Highlights for Anime-Style Food Rendering by Fluid Simulation

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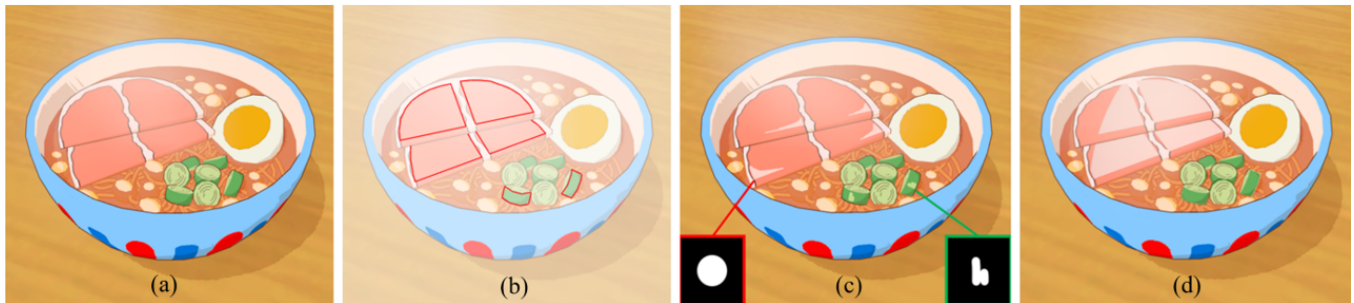


Figure 1: (a) Input 3D models, (b) selected areas to highlight using triangle meshes, (c) our result and input highlight textures, (d) result of applying Blinn's specular model to the same selected areas.

CCS Concepts

• **Computing methodologies** → *Non-photorealistic rendering*.

Keywords

stylized rendering, fluid simulation

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1 INTRODUCTION

We propose a stylization method for highlights in anime-style rendering, mainly for food. In hand-drawn animation, highlights are crucial in making food appear more delicious. Therefore, they are drawn as an artistic expression rather than bound by physical lighting conditions. We focus on the squashed highlight shapes that appear near the edges of food objects. We address the issues where highlights fail to maintain their stylized appearance on flat surfaces in physically-based stylization.

In hand-drawn animations and illustrations, it is common to see highlights being distorted along the edges or corners of surfaces (Fig. 1c). However, reproducing this phenomenon is challenging with Phong or Blinn's specular models, which strongly depend on mesh normals or with environment mapping techniques that

depend on input textures. Thus, we assume that the deformation and movement of highlights in anime-style resemble fluid behavior and simulate their deformation caused by edges using a particle-based fluid representation.

In addition, the shape of the highlights from the physics simulation that appear in the 3D model depends on the mesh normal, so flat surfaces are more likely to be highlighted entirely or not at all. In the work of Anjyo [Anjyo and Hiramitsu 2003], they added parameters based on the tangent space of the mesh in the process of calculating the specular reflection component of Blinn's model to allow deformation, translation, rotation, etc. of the highlights. However, it is difficult to deform on flat surfaces because it depends on the shape of Blinn's model calculation results.

Baking, which involves fixing the lighting results onto a texture, is used to maintain the style of highlights. This approach is less influenced by the mesh normals and can preserve the input highlight style; however, the highlights become fixed to the object and do not deform.

Another method, which provides a representation by blending physical highlights with input shapes, applies an environment mapping generated from highlight shapes to the hair [Tanaka et al. 2014]. However, because it uses environment mapping, it cannot handle changes in highlight shape near the mesh edges.

2 METHOD

Our method reproduces hand-drawn animation-like highlight expressions by combining highlight position calculated using Blinn's model, deformation using fluid simulation, and texture mapping. The fluid simulation is based on the Smoothed Particle Hydrodynamics (SPH) method [Gingold and Monaghan 1977] in UV space.

The overview of our method is as follows:

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- (1) Initial setting, including inputting highlight textures and 3D models (Fig. 2).
- (2) Target position detection using the specular reflection component (Fig. 3a).
- (3) Highlight shape creation using fluid computation (Fig. 3b).
- (4) Rendering the simulation results.

Note that steps 2 to 4 are performed every frame.

2.1 Initial Setting

First, the input highlight styles are black-and-white (background and foreground) binary images (Fig. 2a). Particles are placed in white areas in the image at a constant density using the Poisson Disk Sampling (Fig. 2b). The initial coordinates of each particle are given as I_i , where $i = 1, 2, \dots, n$ and n is the number of particles. Then, triangular meshes of the input model are selected as highlight areas and flattened to UV space for fluid simulation (Fig. 2c).

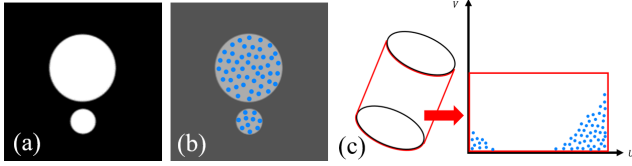


Figure 2: (a) Highlight texture, (b) particle placement with Poisson Disk Sampling, (c) selected mesh and simulation area.

2.2 Calculation of Highlight Position and Shape

Our method calculates the target position to be followed by particles. In the SPH framework, we add the force toward the edges to squash a highlight shape.

2.2.1 Force to move to the target position.

We apply a force to the particles in every frame toward its initial position relative to the target position. When simulating the deformation of highlights squashed toward edges and corners, the highlight texture should be moved more outside the edge to allow particles to escape outside the highlight so that the input highlight shape and the collapsed shape do not overlap too much. Therefore, the target position T^{ext} is represented using position T with the maximum specular reflection component as follows (Fig. 3a):

$$T^{ext} = s(T - M) + M$$

where M is the center of the simulation area (= coordinates averaged over the vertices of the edge) and s is the magnification rate. Then, the force F_i^{ext} to be applied is,

$$F_i^{ext} = k^{ext} \text{Normalize}(T^{ext} + I_i - p_i)$$

where k^{ext} is a constant and p_i is the particle position in the previous frame. I_i is an initial position of each particle, with the texture's center as the origin.

2.2.2 Force to move toward the neighborhood edges (Fig. 3b).

We deform the highlights around the edges (Fig. 1c). We give each particle an external force F_i^{def} as follows:

$$F_i^{def} = \exp(k^{def} \frac{d_1}{d_1 + d_2}) \hat{f}_1 + \exp(k^{def} \frac{d_2}{d_1 + d_2}) \hat{f}_2$$

Note that $\hat{f}_1$ and $\hat{f}_2$ are the normals of the two edges V_0 , the vertex on all edges closest to the particle. k^{def} is a negative constant, and d_1 and d_2 are the shortest distances to each edge.

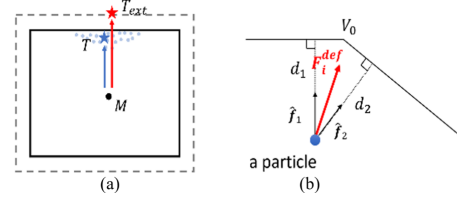


Figure 3: (a) Expand the target position in the outward direction of the area, (b) force to move to target position.

2.3 Rendering

A highlight texture is mapped to the target position, and the outer particles from the mapped highlight are rendered by metaballs (Fig. 4a). Texture mapping hides the unevenness of the highlight appearance due to particle placement and preserves the texture of the undeformed areas (Fig. 4b). The simulation results are then texture mapped to the 3D model to obtain the final output.

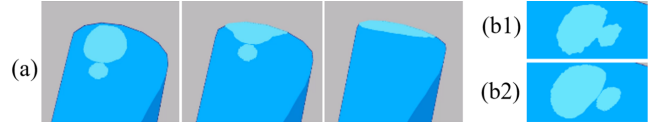


Figure 4: Rendering results (a), comparison between metaballs only (b1) and with texture mapping (b2).

3 RESULTS AND CONCLUSION

Figure 1 compares the results of our proposed method with those physically calculated using Blinn's model. The position of light sources, cameras, and objects in the scene causes highlights to move and deform in real-time.

Our approach has some potential applications. For example, it can be applied not only to food, but also to the eyes and hair of anime-characters. However, because we simulate in UV space, meshes with complex geometry are prone to distortion of the highlight shape. Besides, more than expanding the target position based on the area's center is required. Resolving these issues will make this method more versatile.

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